

M E R N Solution Requirements

shopEZ — B2C E-Commerce Marketplace (MERN Stack)

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Solution Requirements

1. Introduction

This document specifies the complete set of Functional and Non-Functional Requirements for the ShopEZ e-commerce platform. Each requirement is assigned a unique identifier to enable traceability from inception through design, implementation, and User Acceptance Testing (UAT).

2. Functional Requirements

FR-01 to FR-05 — User Management

Req. ID	Requirement Description	Priority	USN
FR-01	The system shall permit new users to register by providing a unique email, full name, and password. Email format validation is enforced.	Must Have	USN-1
FR-02	The system shall authenticate registered users via email and password credentials and issue a signed JWT token upon success.	Must Have	USN-2
FR-03	The system shall store JWT tokens exclusively in HTTP-only cookies to prevent client-side JavaScript access (XSS mitigation).	Must Have	USN-2
FR-04	The system shall enforce Role-Based Access Control (RBAC) distinguishing between the customer, seller, and admin roles.	Must Have	USN-0
FR-05	Authenticated users shall be able to update their profile including name, password, and saved shipping addresses.	Should Have	USN-1

3. Non-Functional Requirements

Req. ID	Category	Requirement Description	Priority
NFR-01	Security	All user passwords must be hashed using Bcrypt (salt rounds >= 10) prior to database persistence.	Must Have
NFR-02	Security	All API routes that modify data must require a valid JWT; unauthenticated requests shall return HTTP 401.	Must Have
NFR-03	Security	Admin-only routes shall apply an additional isAdmin middleware check; non-admin access returns HTTP 403.	Must Have
NFR-04	Performance	Product catalogue API responses must be returned within 500ms under normal load conditions.	Should Have
NFR-05	Performance	All product images must be served via the Cloudinary CDN to ensure sub-second image load times globally.	Should Have
NFR-06	Usability	The frontend must be fully responsive from 320px (mobile) to 1920px (desktop).	Must Have
NFR-07	Reliability	MongoDB's document atomicity must ensure stock deduction and order creation are performed consistently, preventing overselling.	Must Have
NFR-08	Maintainability	All backend code must follow clean architecture: Router -> Middleware -> Controller -> Model. No business logic in routes.	Should Have
NFR-09	Scalability	The stateless JWT model shall ensure the API can be horizontally scaled without shared session state.	Could Have

4. Requirements Traceability Matrix

Req. ID	Design Artefact	Implementation File	UAT Test Case
FR-01, FR-02, FR-03	Solution Architecture	authRoutes.js, authController.js	UAT-001, UAT-002, UAT-003
FR-06, FR-07, FR-08	Data Flow Diagram	productRoutes.js, productController.js	UAT-005, UAT-006
FR-11, FR-12	ER Diagram	cartRoutes.js, cartController.js, Cart.js model	UAT-008, UAT-009

Req. ID	Design Artefact	Implementation File	UAT Test Case
FR-13, FR-14, FR-15	User Flow Sequence	orderRoutes.js, orderController.js	UAT-011, UAT-012
FR-10	Solution Architecture	sellerRoutes.js, adminRoutes.js	UAT-013, UAT-014
NFR-01, NFR-02	Solution Architecture	authMiddleware.js	UAT-004, UAT-016

